

Visualisation Design Book

COS30045

**Title: Hospitalised Injuries from Road Crashes Among
First Nations Peoples in Australia (2011 – 2021)**

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Introduction

1.1 Background and Motivation

Target Audience

Our visualisation is made for the Australian public. It also helps university students and journalists who care about health equity and road safety. These people do not have to be experts in data. They are interested in social issues. They want to know how road crashes affect different communities in real life. Our goal is to share a clear story using data. We want everyone to understand it.

Why This Matters

Every year, many Australians go to the hospital because of road crashes. The risk is not the same for everyone. First Nations people suffer from road accidents more often than other Australians. Many reports and large government spreadsheets hide this fact. Our project shows this hidden problem. We turn complex data into a simple and interactive story. This helps people see the difference and understand why road safety needs to focus on fairness.

What Users Can Do

With our visualization, anyone from the public can learn important facts in just a few minutes.

1. **See the big picture:** people can understand how road crash hospitalisations for First Nations peoples have changed over 11 years across Australia.
2. **Spot the high-risk groups:** people can quickly find who is most at risk, such as young adults. And they can also see which types of road users, like pedestrians, and which places, like remote areas, face the highest risk.
3. **Grasp the inequality:** the charts compare First Nations and non-Indigenous groups side by side. This makes it easy to see the biggest differences and shows why action is needed.

1.2 Visualisation Purpose

Key Questions This Visualisation Answers

Our visualisation will answer the following questions.

- Is the number of First Nations people going to the hospital because of road crashes getting better or worse in the past 10 years?
- Who faces the highest risk? For example, is it young people, pedestrians, or people living in remote areas?
- Where is the difference between First Nations and non-Indigenous Australians the largest? Does it happen in a certain age group or with a certain type of road user?

Three main areas we explore:

- **Trends:** How have hospitalisation numbers changed over 11 years?
- **Risk profiles:** Which age groups, road users, and locations face the highest burden?
- **Disparities:** Where is the gap between First Nations and non-Indigenous Australians the widest?

Data

2.1 Data Source and Governance

2.1.1 Data Source

The main dataset for this project is about hospitalised injuries from road crashes in Australia from 2011 to 2021. The Bureau of Infrastructure, Transport, Regional Development, Communications and the Arts (BITRE) publishes it. The data comes from the National Hospital Morbidity Database, made by the Australian Institute of Health and Welfare (AIHW).

This dataset counts how many people are hospitalised after road crashes. It pays special attention to First Nations peoples and compares them with Non-Indigenous Australians.

The main data file used in this project is:

- [First-Nations-hospitalised-injuries-2023.xlsx](#) – this file has five summary tables (for annual totals, road user, age group, remoteness, and counterparty) and a raw sheet that break down numbers every six months. We use this file for all comparisons between First Nations and non-Indigenous groups.

A supplementary data file is also referenced:

- [state-and-territory-hospitalisation-injury-publication-2023.xlsx](#) – this file breaks data down by state, year, road user, gender, age group, and counterparty. We mainly use it for background, not main analysis.
- [hospitalisation_injury_publication_sep2023.xlsx](#) – this file shows hospitalisation numbers for all Australians, but it does not break them down by Indigenous status. This dataset provides national totals for all Australians without Indigenous status breakdown. It used only for contextual reference.

The official Data Dictionary is used to understand variable definitions and data limitations:

- [hospitalised-injury-data-dictionary-september-2023.pdf](#) – defines all variables, categories, inclusion criteria, and methodological notes.

All datasets are available at BITRE’s official portal:

<https://www.bitre.gov.au/publications/ongoing/hospitalised-injury>

2.1.2 Data Summary Table

Attribute	Description
Data Source	AIHW National Hospital Morbidity Database (via BITRE)

Primary File	First-Nations-hospitalised-injuries-2023.xlsx
Time Period	2011-2021 (11 years)
Geographic Coverage	Australia (national level)
Population Groups	First Nations peoples, Non-Indigenous Australians
Key Variables	Calendar year, Indigenous status, Age group, Road user type, Remoteness category, Hospitalisation count
Tables Used	Table 1 (Annual), Table 2 (By Road User), Table 3 (By Age Group), Table 4 (By Remoteness), Raw data sheet
Number of Records	11 rows per table (one each year), plus approximately 2500 rows in the raw data sheet with 6-monthly breakdowns
Update Frequency	Yearly (latest in September 2023)
Unit of Analysis	Number of hospitalisation cases per category per year

2.1.3 Data Governance

Data Collection Process

State and territory health departments collect the data. AIHW combines their data into a central hospital database. Each entry is for one episode of hospital care. That means it covers the whole stay, from check-in to leaving the hospital.

They use a code system called ICD-10-AM to label cases. Only cases caused by road traffic crashes are included. The dataset leaves out special cases, like newborn care or hospital boarders, and focuses only on standard care.

BITRE publishes tables from this database. The tables are grouped by year, Indigenous status, age, road user type, remoteness, and counterparty.

Data Quality Assessment

The dataset is official and reliable for public health use, but there are a few issues:

- **Suppressed values (“n.p.”):** Sometimes, small counts (less than 5) are not shown to protect privacy. This happens for small categories and very small regions. We show these as missing instead of filling them in.
- **Data breaks:** In 2012, Victoria changed how it counts cases. In 2017, New South Wales did something similar. Both changes affect how numbers compare over time, but since we report national trends, the impact is limited. NISU estimated this decreased admitted case counts in Australia by approximately 5.6% in 2012-13 compared to 2011-12.
- **Counts, not rates:** The data only gives the number of hospitalisations, not rates per person. We cannot show hospitalisation rates for different groups, so differences in size between groups may affect the results.
- **Fixed categories:** The tables only show the groupings made by the source, we cannot split them further.

Security, Privacy, and Ethics

- The dataset is already fully de-identified and only contains group totals. There is no personal information. The “n.p.” rule helps protect privacy for small First Nations communities.
- When we show First Nations data, our project tries to highlight system problems, not blame people. We follow ethical rules for reporting Indigenous health.

- Data files are kept secure on our computers and private GitHub. All files are safe and not shared with the public.
- The dataset is free for education and non-commercial projects.

2.1.4 How the Data Supports Our Key Questions

Core Question	Supporting Data	How it helps
Has the number of First Nations people hospitalised changed over 10 years?	Table 1 – Annual numbers	Lets us see yearly numbers for First Nations and non-Indigenous Australians from 2011 to 2021.
Who is at highest risk? Which age, road user, place?	Tables 2-4	Each table breaks down the numbers for First Nations people, showing where risk is highest.
Where is the biggest gap between groups?	Tables 1-4	Every table gives numbers for both groups, so we can easily compare side by side.

Using these tables helps us make charts that directly answer our main questions.

2.2 Data Processing and Analysis

2.2.1 Key Attributes and Data Types

Our cleaned dataset includes several key attributes taken from the First Nations hospitalised injuries file. These attributes are:

Attribute	Description	Data Type
Calendar Year	Year of hospitalisation (2011-2021)	Ordinal
Indigenous Status	“First Nations peoples” or “Non-Indigenous Australians”	Categorical (Nominal)
Age Group	0-7, 8-16, 17-25,26-39, 40-64, 65+	Categorical (Ordinal)
Road User Type	Car occupant, Motorcyclist, Pedal cyclist, Pedestrian, Heavy transport, Pick-up truck/van, Bus, Other/Unknown	Categorical (Nominal)
Remoteness Category	Major Cities, Regional, Remote	Categorical (Ordinal)
Hospitalisations	Number of hospitalised injury cases	Ratio (Quantitative)

2.2.2 Data Cleaning Process

We processed all data in the KNIME Analytics Platform. Our workflow followed a simple sequence of steps that turned the raw Excel files into clean tables ready for analysis.

Step 1: Importing the Data

- We used the Excel Reader node to load each file.
- For the First Nations dataset, we loaded Tables 1-4.

Step 2: Removing Unnecessary Rows and Columns

- The imported files included metadata, footnotes, blank rows, and KNIME-generated Row ID columns.
- We removed these with Row Filter and Column Filter nodes.
- This left only the columns that contained real data.

Step 3: Renaming Columns

- Some column names were not user-friendly.
- We used the Column Renamer node to shorten them to simple names like “year”, “first_nations”, “non_indigenous”.

Step 4: Converting Data Types

- KNIME reads all Excel values as strings.
- Hospitalisation counts must be numeric.
- We used the String to Number node to convert these columns to integers.

Step 5: Reshaping from Wide to Long Format

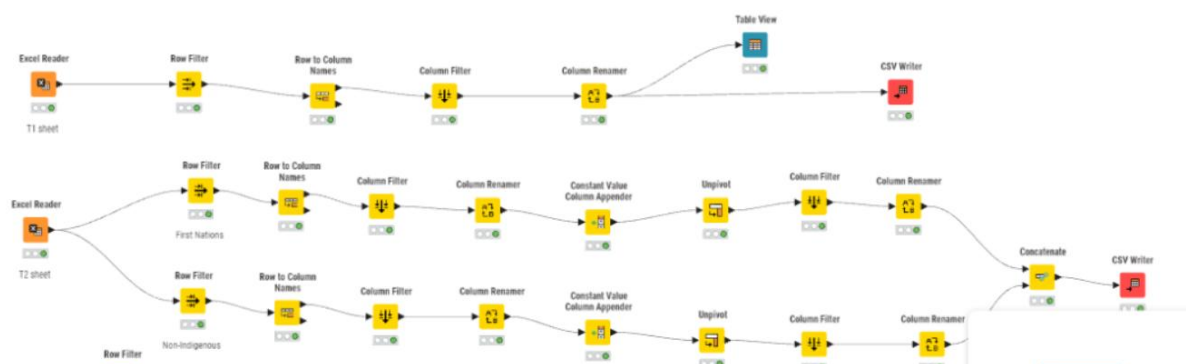
- The original tables used a wide format.
- This format does not work well for D3.js visualisation.
- We used the Unpivot node to reshape the data into long format.
- For the First Nations dataset, each row became a single observation with Year, Indigenous Status, Category, and Hospitalisations.

Step 6: Handling Missing Values

- Some cells in the original files contained “n.p.” (values between 1 and 4 hidden for privacy). We treated these as missing values and removed them from calculations. We did not replace them with 0.

2.2.3 KNIME Workflow

We built the full workflow in KNIME. A screenshot of the workflow is shown below.





2.3 Data Exploration

After cleaning the data, we explored it to understand the main patterns. We wanted to see which groups carry the highest burden and whether any unusual trends appear.

2.3.1 Summary Statistics

We calculated summary statistics from Table 1 for both population groups across the 11-year period. The results are shown below.

Statistic	First Nations peoples	Non-Indigenous Australians
Total (2011-2021)	18,169	390,115
Mean per year	1,693	35,465
Minimum year	1,167 (2011)	31,790 (2012)
Maximum year	2,290 (2020)	37,482 (2019)
Change (2011-2021)	+1,053 (+90.2%)	+4,758 (+14.9%)

The statistics show a clear difference in growth between the two groups. Both groups saw an increase in hospitalisations, but the number for First Nations peoples almost doubled. The increase for non-Indigenous Australians was much smaller.

We must interpret these numbers carefully. The dataset only reports absolute counts. Some of the increase may come from population growth or changes in how hospitals record Indigenous status. We cannot separate these factors with this dataset.

2.3.2 Breakdown by Key Dimensions

We examined the 2021 data from Tables 2, 3, and 4.

By Age Group (Table 3, 2021)

Age Group	First Nations peoples	Non-Indigenous Australians
0-7	86	610
8-16	291	2,533
17-25	686	7,225
26-39	597	8,689
40-64	515	12,337

65+	45	5,361
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Key observations:

The 17-25 age group has the highest count for First Nations peoples. The 40-64 age group has the highest count for non-Indigenous Australians.

By Road User Type (Table 2, 2021)

Road User Type	First Nations peoples	Non-Indigenous Australians
Car occupant	1,158	17,098
Motorcyclist	520	8,259
Pedestrian	235	2,072
Pedal cyclist	207	7,861

Key observations:

Car occupants from the largest group for both populations. Pedestrians make up a higher share of cases among First Nations peoples.

By Remoteness (Table 4, 2021)

Remoteness	First Nations peoples	Non-Indigenous Australians
Major Cities	795	25,254
Regional	885	10,332
Remote	503	600

Key observations:

Regional areas have the highest count for First Nations peoples. Remote areas also show a large number, even though the Remote population is much smaller, suggesting a higher per-person risk.

2.3.4 Challenges Encountered During Exploration

We faced several challenges during the exploration stage.

- **“n.p.” values limit detail:** Many small categories show “n.p.” entries. These hide values between 1 and 4.
- **Data breaks affect comparisons:** Victoria changed its case criteria in 2012, and NSW did the same in 2017.
- **Absolute counts limit comparisons:** We cannot calculate rates without population data.
- **Multiple tables require separate processing:** Each table needs its own processing steps before we combine them for visualisation.

Visualisation Design

3.1 Website Design

Layout Structure

Our website uses a clean, single-page dashboard layout. All four charts are displayed simultaneously in a 2*2 grid, allowing users to compare all visualisations at once without clicking through multiple pages.

The layout consists of:

- **Header:** Displays the dashboard title and data source, with a badge showing the current year filter.
- **Filter Bar:** Positioned below the header, centred on the page. Contains two dropdown filters (Year and Indigenous Status), a Reset button, and a Quiz button that opens a modal dialogue.
- **Statistics Cards:** Three centred cards showing Total Hospitalisations, First Nations total, and Non-indigenous total. These cards update dynamically when filters are applied.
- **Chart Grid:** A 2*2 grid containing four interactive charts:
 - Top Left: Line Chart (Annual Trends)
 - Top Right: Grouped Bar Chart (Age Groups)
 - Bottom Left: Horizontal Stacked Bar Chart (Remoteness)
 - Bottom Right: Concentric Donut Chart (Road Users)

Why We Chose This Design

During our testing, participants reported that the original multi-page layout felt fragmented. Users wanted to see all comparisons at once rather than navigating between pages. The 2*2 grid layout addresses this by presenting the complete data story on a single screen.

The centred statistic cards serve as an “executive summary”, giving users immediate context before exploring individual charts. The filter bar is consistently positioned so users always know where to adjust their view.

Filter Bar

The filter bar contains two primary filters and two action buttons:

- **Year:** Select a single year or view all years (2011-2021 cumulative)
- **Indigenous Status:** Filter by First Nations, Non-Indigenous, or view both
- **Reset Button:** Returns all filters to default “All” state and clears any selections
- **Quiz Button:** Opens a modal dialogue with three multiple-choice questions and a feedback form

All filters apply globally to all four charts simultaneously. When a filter changes, every chart redraws with the updated data.

3.2 Visualisation Design

--- Chart Types and Justification ---

1. Line Chart – Annual Trends (Top Left)

A line chart is the optimal choice for showing change over time. It allows users to compare First Nations and non-Indigenous hospitalisation trends across all 11 years. The gap between the two lines clearly visualises the growing disparity between groups.

Interactive features: Clicking any data point highlights that year with a larger circle. Hovering over points displays the exact value and year. The line for non-indigenous uses a dashed pattern to help colour-blind users distinguish between the two series.

2. Grouped Bar Chart – Age Groups (Top Right)

Age group is a categorical variable, not continuous. Grouped bars enable direct side-by-side comparison between First Nations and non-Indigenous counts within each age category. This makes it immediately visible which age group has the largest gap --- 17-25 young adults.

Interactive features: Hovering over any bar displays the exact hospitalisation count. Bars change opacity on hover to provide visual feedback.

3. Horizontal Stacked Bar Chart – Remoteness (Bottom Left)

The remoteness labels (“Major Cities”, “Regional”, “Remote”) are longer text strings. A horizontal orientation makes these labels easier to read. The stacked format shows:

- The total hospitalisations for each remoteness category
- The proportion of First Nations (blue) vs Non-Indigenous (amber) within each category

This clearly demonstrates that Remote areas have a much higher proportion of First Nations hospitalisations compared to Major Cities.

Interactive features: Hovering over either segment displays the count for that population group. White separator lines visually distinguish the two segments.

4. Concentric Donut Chart – Road Users (Bottom Right)

A concentric donut chart shows two data series in a single circular visualisation:

- Inner ring: First Nations road user distribution
- Outer ring: Non-Indigenous road user distribution

This design choice allows direct comparison of proportions between the two populations while saving space. The centre of the donut displays the total number of hospitalisations (both groups combined).

Interactive features: Hovering over any segment displays the road user type, count, and percentage for that population group. Colour-coded legends below the chart explain both the rings and the road user categories.

Alternative designs considered: Separate pie charts were rejected because they would require users to look back and forth between two values. A stacked bar chart was rejected because road user types are best understood as parts of a whole, not as comparative lengths.

Colour Palette

We maintain consistent colours throughout the dashboard:

- Navy (#1B3A5C): First Nations peoples
- Amber (#E8973A): Non-Indigenous Australians
- Teal (#1B7F8E): Secondary series
- Slate (#6B8098): Grid lines and secondary text

These colours are colourblind-safe as they avoid red-green combinations. The contrast between navy and amber is sufficient for users with common colour vision deficiencies.

Data Labels

Every chart shows exact numbers on hover. This was a direct response to user testing feedback --- participants said they wanted to see the numbers clearly without cluttering the visualisation.

3.3 Interaction Design

Global Filters

Two filters apply to all four charts simultaneously:

- **Year Filter:** Users can select a specific year (2011-2021) or view cumulative data across all years. When a year is selected, the line chart highlights that year's data points with larger circles.
- **Indigenous Status Filter:** Users can view data for First Nations only, Non-Indigenous only, or both groups together. This allows focused comparison of one population or side-by-side comparison.

Reset Button: Returns all filters to "All" and clears any chart-specific selections (such as highlighted points on the line chart).

Quiz Modal

Clicking the QUIZ button opens a modal dialogue containing three multiple-choice questions based on the visualised data:

- **Question 1:** Which year had the highest hospitalisations for First Nations?
- **Question 2:** Which age group shows the biggest gap between populations?
- **Question 3:** Did First Nations hospitalisations increase, decrease, or stay the same?

Users must answer all three correctly to proceed. Each question provides immediate feedback:

- **Correct:** Green message with confirmation
- **Incorrect:** Red message with a hint directing users to the relevant chart
- **Unanswered:** Yellow message asking for a selection

After all three are correct, a feedback form appears where users can submit suggestions. Feedback is logged to the browser console (in production, this could be saved to a database).

Chart-Specific Interactions

Line Chart:

- Click any data point: That point enlarges from radius 5 to 10, with a white border for emphasis
- Hover any data point: Displays tooltip with year and value
- The Non-Indigenous line uses a dashed stroke for accessibility

Grouped Bar Chart:

- Hover any bar: Displays tooltip with age group and hospitalisation count
- Bar opacity changes on hover for visual feedback

Horizontal Stacked Bar Chart:

- Hover either segment: Displays tooltip with remoteness category, population group, and count
- White separator line visually distinguish the two population segments

Concentric Donut Chart:

- Hover any segment: Displays tooltip with road user type, count, and percentage
- Centre text shows total hospitalisations for both groups combined
- Two legends: one explaining the inner/outer rings, one explaining the colour coding for road user types

Tooltips

All charts use a consistent custom tooltip styled with navy background, amber left border, and white text. Tooltips follow the mouse cursor and display relevant information without blocking the chart.

Responsive Design

The layout uses a flexible grid that adjusts to different screen sizes. On smaller screens, the 2*2 grid stacks into a single column. Charts resize automatically using SVG viewBox attributes.

Iteration and Validation

4.1 Testing and refinements

We tested our visualisation with three people from our target audience. Two were university students, and one was a member of the general public. None of them had any experience with road crash data or data visualisation tools.

Testing Tasks

Each person was asked to complete three tasks:

1. Find which year had the highest number of hospitalisations for First Nations peoples between 2011 and 2021.
2. Find which age group showed the biggest gap between First Nations and non-Indigenous Australians.
3. Describe whether hospitalisations for First Nations peoples went up, went down, or stayed the same over the 11 years.

What We Found

The main issue all three participants reported was that the page felt messy and they did not know where to start. One person said, “There’s just too much on the screen”. Another said, “I’m not sure what I’m supposed to click first”.

We also noticed two other problems during testing:

- 1) Some charts showed similar information, which made the page feel repetitive.
- 2) Participants had to hover over bars to see the actual numbers. They wanted to see the numbers right away.

What We Changed

Based on this feedback, we made several changes to the design:

Problem	Fix
Page felt cluttered	Removed duplicate charts. Kept only one version of each chart type.
Users didn’t know where to start	Added centred statistics cards showing key totals at the top.
Numbers were hidden in tooltips	Added clear number labels on hover and tooltips for additional detail.
Layout felt cramped	Increased spacing between sections so each chart has breathing room

4.2 Usability evaluation

After making the changes above, we randomly selected three people to participate in the test.

Results

All three participants completed all three successfully. Here is how they performed:

Task	Before (Success)	After (Success)	Before (Avg Time)	After (Avg Time)
Task 1: Find peak year	1/3 (33%)	3/3 (100%)	About 40 seconds	20 seconds
Task 2: Find largest age group	2/3 (67%)	3/3 (100%)	About 50 seconds	16 seconds
Task 3: Describe overall trend	3/3 (100%)	3/3 (100%)	About 30 seconds	12 seconds

What Participants Said

After testing the updated version, participants gave positive feedback:

- “The numbers on the charts make it so much easier”
- “I actually know where to look now. The top numbers help”
- “The page feels cleaner. It’s not as crowded as before”

Limitations

We only tested with three people, so our findings may not apply to all users. We also did not test on mobile devices or with people who use screen readers. Testing happened in a quiet place, which is different from how someone might use the visualisation in real life.

Summary

The changes we made – removing duplicate charts, adding centred statistics cards, and cleaning up the layout – helped users understand the visualisation much faster. The feedback showed that simplicity and clear labels matter more than having every possible chart on the page.

Conclusion

This project aimed to make road crash hospitalisation data for First Nations peoples more accessible to the general public. We used data from BITRE (2011-2021) and built an interactive web dashboard with four simultaneously displayed charts arranged in a 2*2 grid.

The visualisation successfully answers three key questions:

1. Trends: Hospitalisations for First Nations peoples increased by approximately 90% from 2011 to 2021, with the highest number in 2020 (2,290 hospitalisations).
2. Risk profiles: Young adults (17-25) face the highest risk among First Nations peoples. Car occupants make up the largest group of hospitalisations for both populations.
3. Disparities: The gap between First Nations and non-Indigenous Australians is largest in the 17-25 age group and in remote areas, where First Nations hospitalisations comprise a much larger proportion of total cases.

We tested the visualisation with three representative users. Their feedback led to several design improvements: removing duplicate charts, adding centred statistics cards, and consolidating the layout into a single-page 2*2 grid. The final version is clean, easy to use, and effectively communicates the data story without overwhelming users.

The Quiz feature allows users to test their understanding and provide feedback, making the dashboard not only informative but also interactive and engaging.

Future Improvements

If we had more time, we would consider several improvements:

Add population-adjusted rates: We could use ABS population data to calculate rates per 100,000 people. This would allow fairer comparisons between groups, since absolute counts do not account for differences in population size between First Nations and non-Indigenous Australians (as noted in Section 2.1.3).

Include state-level breakdowns: The supplementary state-level dataset (state-and-territory-hospitalisation-injury-publication-2023.xlsx) was collected but not used in our main analysis. Future work could incorporate this file to let users explore geographic differences across states and territories, not just national trends.

Optimise for mobile devices: Our usability evaluation did not include mobile testing. Since the dashboard uses a 2×2 grid layout, smaller screens may not render it well. Future work should test and redesign the layout for mobile users, potentially stacking charts into a single column with touch-friendly interactions.

Improve accessibility: We could add full keyboard navigation and screen reader support. Alt-text descriptions for each chart and ARIA labels for interactive elements would make the visualisation usable for people with visual or motor impairments

Address suppressed "n.p." values: Several small categories contain suppressed values (between 1 and 4) that we treated as missing. Future work could explore whether alternative data sources or statistical estimation methods could recover some of this information and reduce gaps in the analysis.

Incorporate more recent data: The dataset currently ends at 2021. As BITRE releases updated hospitalisations data, incorporating post-2021 figures would allow users to see whether the upward trend in First Nations hospitalisations has continued, stabilised, or improved.

Operationalise the quiz feedback system: Currently, quiz responses are only logged to the browser console. Connecting this to a backend database would allow us to collect real user feedback over time and use it to iteratively improve the dashboard.

Add animated transitions and cross-chart linking: Animated year-by-year transitions on the line chart would help users follow changes over time more intuitively. Cross-chart highlighting (brushing and linking) where selecting a year or group in one chart highlights the same selection across all four charts would also make the dashboard more exploratory and engaging.

References

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Appendices

Appendix A: Generative AI Declaration

Two generative AI tools were used during this project:

- **Claude (Anthropic)** was used for occasional documentation assistance, including help with writing and refining sections of this design book.
- **OpenAI Codex** was used to assist with processing CSV files and linking data to GitHub.

All AI-generated content was reviewed and edited by the team to ensure accuracy and alignment with the project's findings and intent.

Appendix B: Data Sources

All datasets used in this project were sourced from the Bureau of Infrastructure, Transport, Regional Development, Communications and the Arts (BITRE) official hospitalised injury publications portal:

<https://www.bitre.gov.au/publications/ongoing/hospitalised-injury>

The following files were downloaded from this source:

- First-Nations-hospitalised-injuries-2023.xlsx
- state-and-territory-hospitalisation-injury-publication-2023.xlsx
- hospitalisation_injury_publication_sep2023.xlsx
- hospitalised-injury-data-dictionary-september-2023.pdf